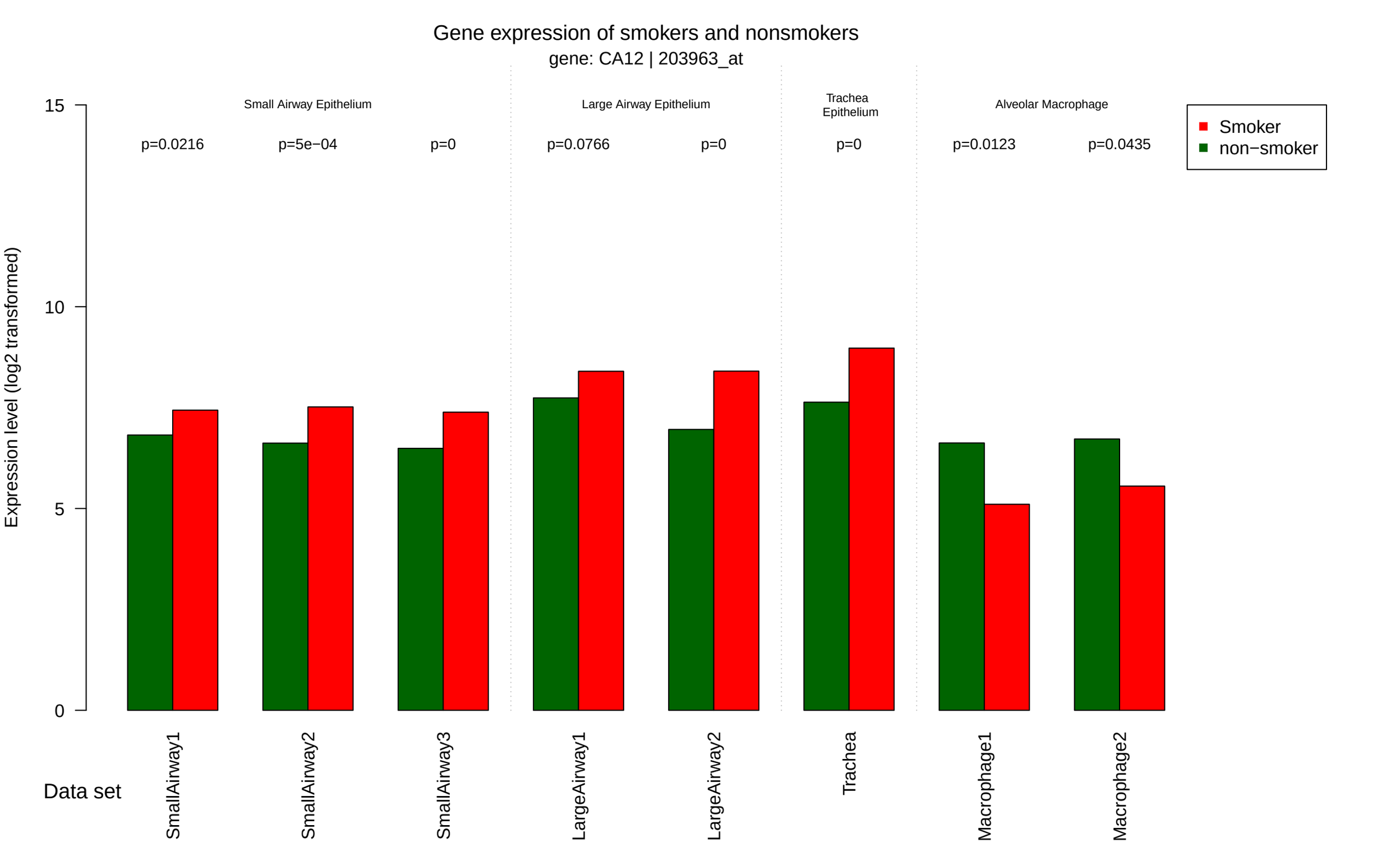
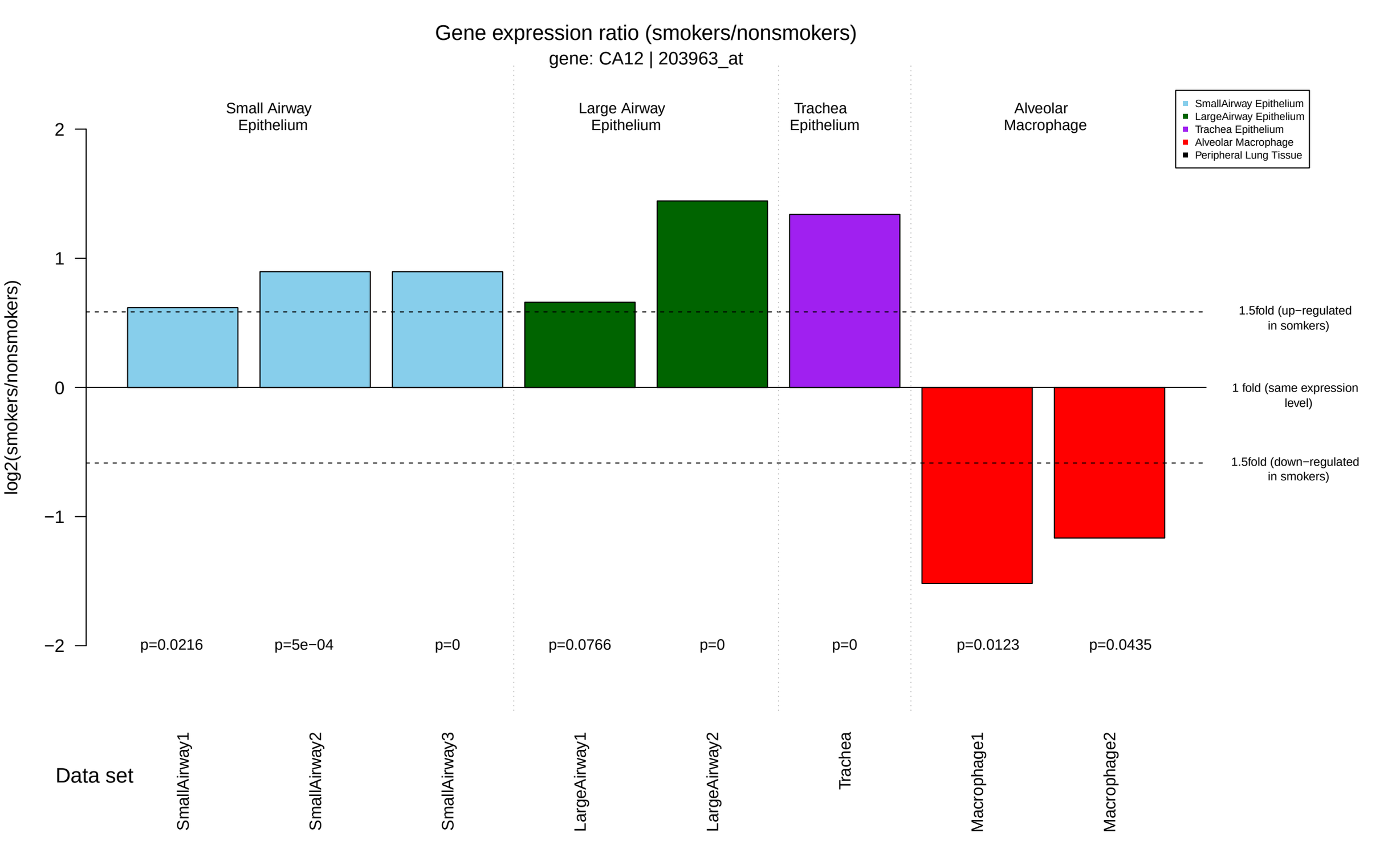
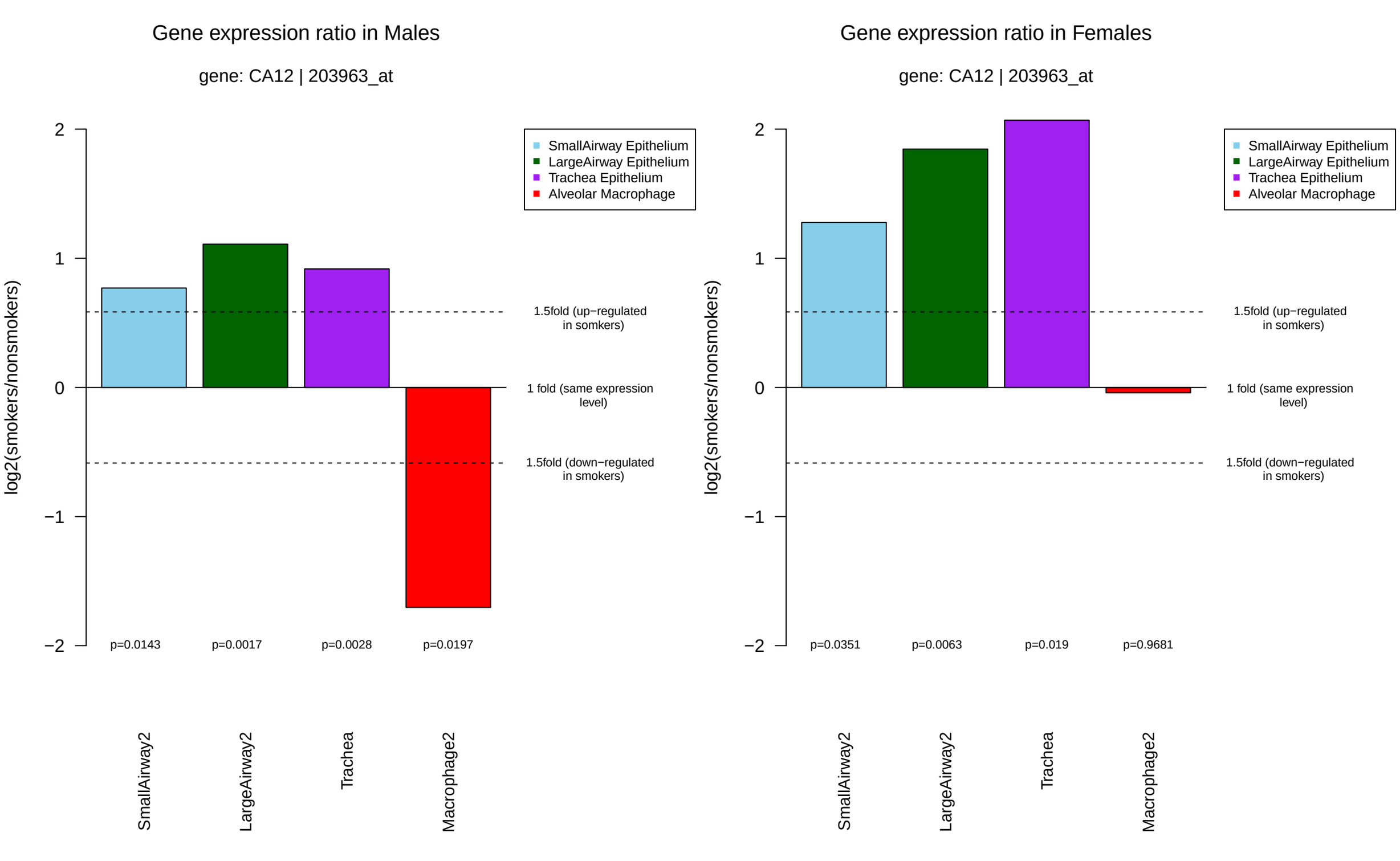
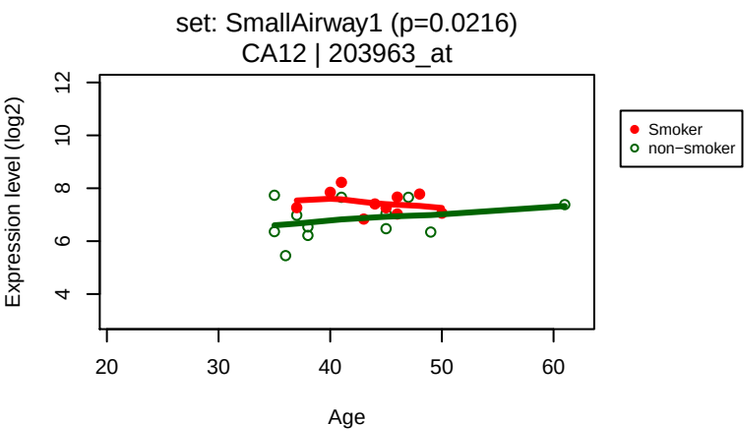
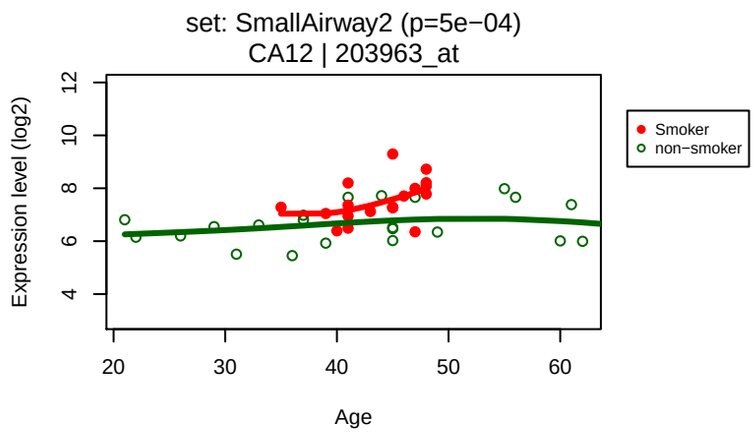
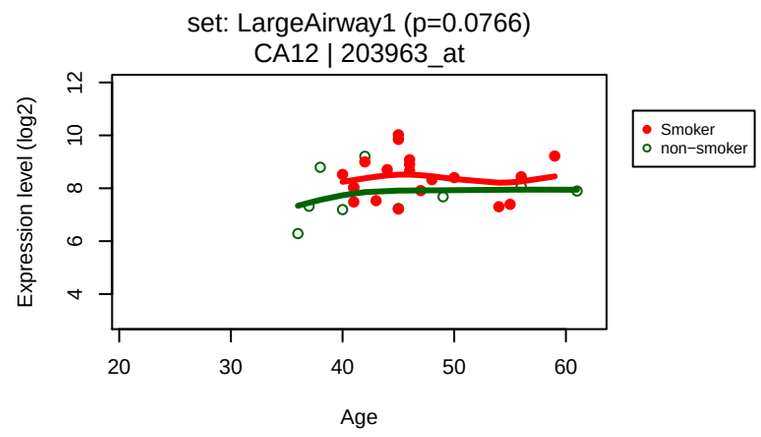

Gene expression of smokers and nonsmokers
gene: CA12 | 203963_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0216
p=5e−04
p=0
p=0.0766
p=0
p=0
p=0.0123
p=0.0435
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

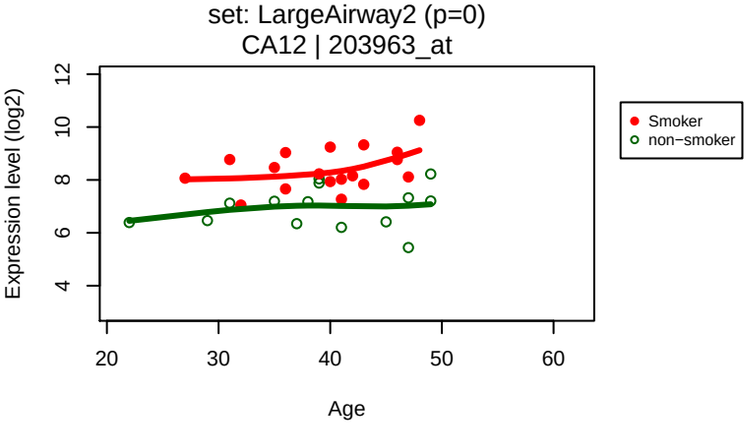

Gene expression ratio (smokers/nonsmokers)
gene: CA12 | 203963_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0216
p=5e−04
p=0
p=0.0766
p=0
p=0
p=0.0123
p=0.0435
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

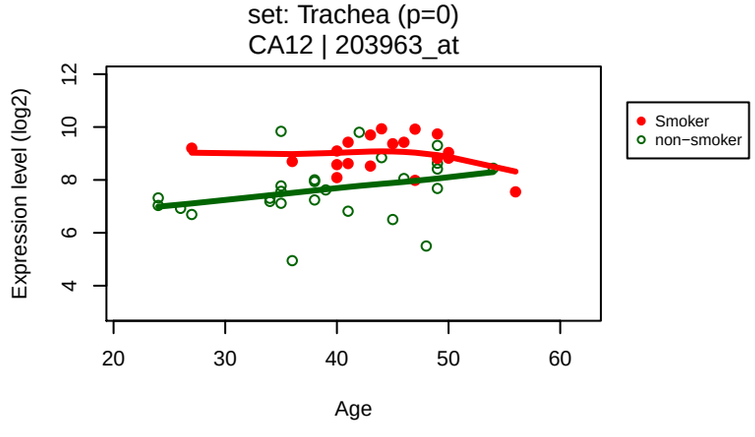

Gene expression ratio in Males
gene: CA12 | 203963_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0143
p=0.0017
p=0.0028
p=0.0197
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CA12 | 203963_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0351
p=0.0063
p=0.019
p=0.9681
SmallAirway2
LargeAirway2
Trachea
Macrophage2

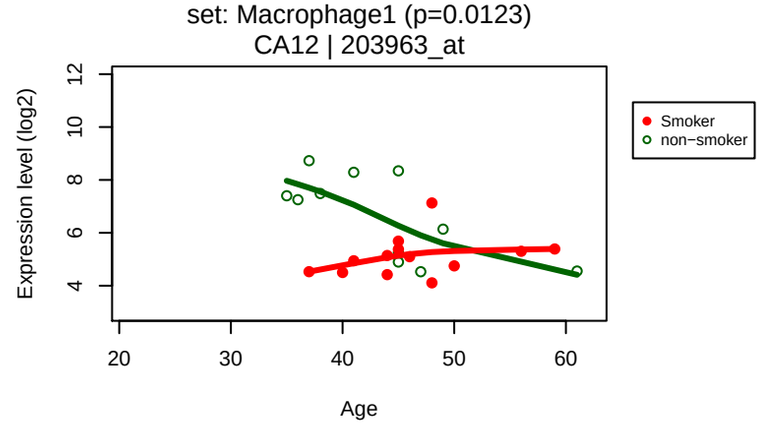

set: SmallAirway1 (p=0.0216)
CA12 | 203963_at
Expression level (log2)
Age
Smoker
non−smoker

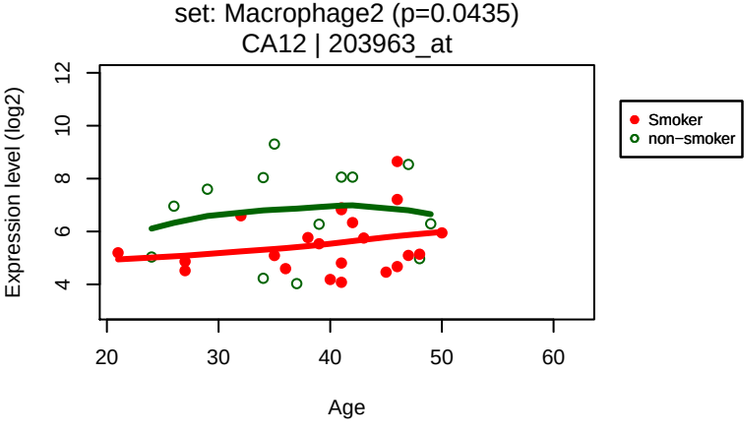

set: SmallAirway2 (p=5e−04)
CA12 | 203963_at
Expression level (log2)
Age
Smoker
non−smoker

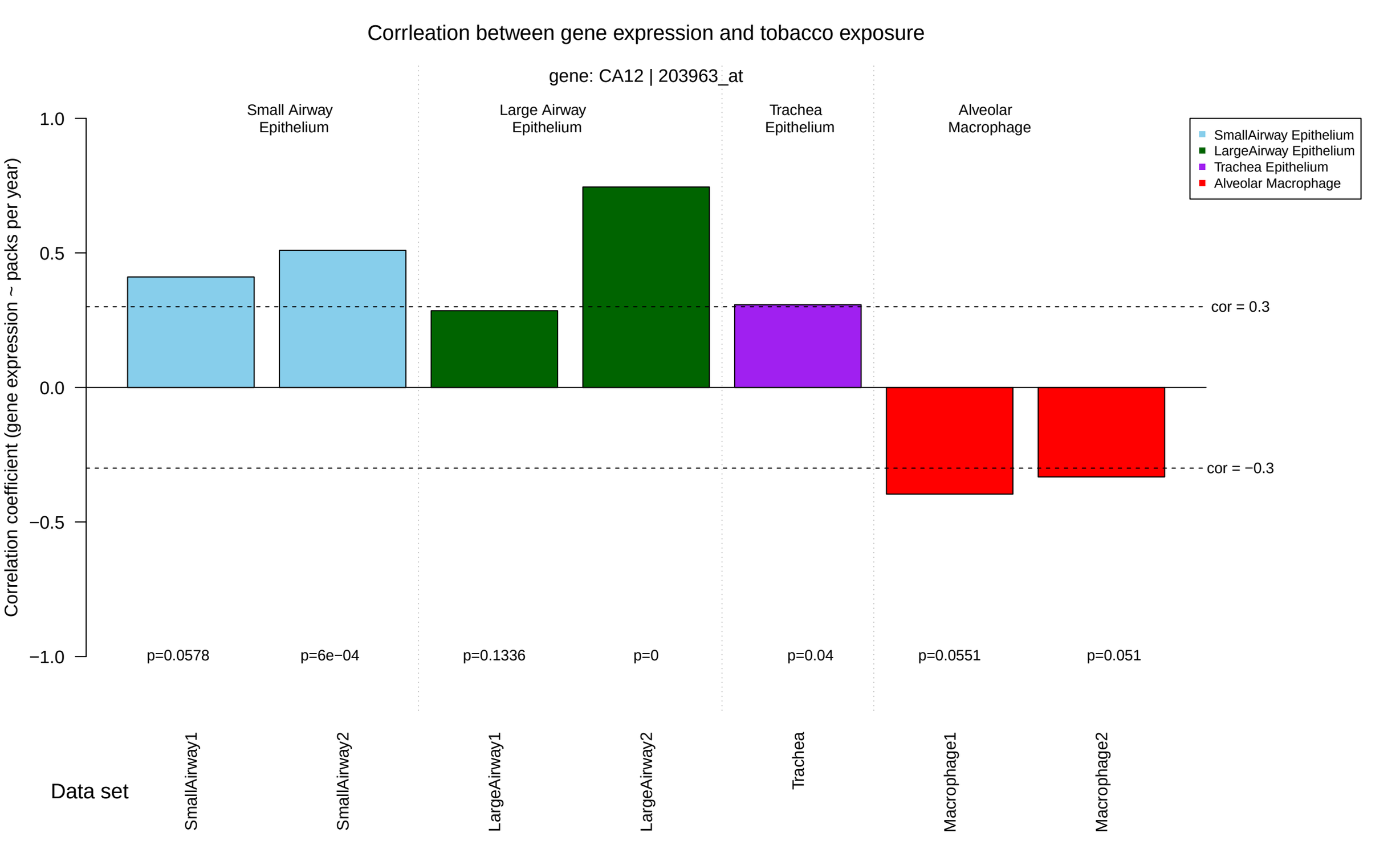

set: LargeAirway1 (p=0.0766)
CA12 | 203963_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0)
CA12 | 203963_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
CA12 | 203963_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0123)
CA12 | 203963_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0435)
CA12 | 203963_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CA12 | 203963_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0578
p=6e−04
p=0.1336
p=0
p=0.04
p=0.0551
p=0.051
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2